Current Microbiology (2024) 81:8
https://doi.org/10.1007/s00284-023-03492-w

Predicting Human Risk with Multidrug Resistant *Enterobacter hormaechei* MS2 having *MCR 9* Gene Isolated from the Feces of Healthy Broiler Through Whole-Genome Sequence-Based Analysis

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Received: 9 April 2023 / Accepted: 14 September 2023 / Published online: 15 November 2023

Abstract

The zoonotic spread of antimicrobial resistance (AMR) and the associated infections are becoming a major threat to the human population worldwide. Strategies to identify the potential pathogen dissemination by seemingly healthy livestock are at a nascent stage and it is of significant importance to monitor environmental evolution of AMR. In this study, a multidrug resistant strain (MDR) of *Enterobacter hormaechei* MS2 isolated from the feces of healthy broiler chicken has been characterized by whole-genome sequencing-based method. Here, the isolate was primarily subjected to antimicrobial susceptibility testing followed genome sequencing and analysis. From the antimicrobial susceptibility testing result, the strain was found to be resistant to multiple classes of drugs including the colistin which is an important last resort drug used to treat infectious diseases. The resistome prediction of genomic data further revealed the presence of 7 perfect and 26 strict hits including those for *MCR-9* and *FosA2*. The pathogenicity prediction has also demonstrated the strain to have the potential to be a human pathogen with 0.72 probability. The phylogenetic analysis has also supported the zoonotic potential of the strain due to its clustering with isolates from both human and livestock-associated host groups. The results of the study suggest the need for a strong surveillance system to identify the opportunistic zoonotic pathogens to prevent a silent AMR menace mediated by them. Carriage of multi-drug resistant strains in the livestock gut microbiome is also a serious concern as it has high AMR transmissibility through contact and supply chain activities.

Introduction

Antimicrobial resistance (AMR) in livestock is one of the major challenges to the currently used therapeutic methods. The unscientific usage and administration of antibiotics and non-antibiotic antimicrobial agents in the livestock sector has significantly contributed to the environmental antimicrobial resistance emergence [1]. One of the highly demanding livestock is the broiler chicken as it produces high quantity of meat within a short time. However, the infections in broiler chickens caused by the MDR pathogens and its treatment challenges significantly affect the farming and profit [2, 3]. The most widely investigated concern for AMR are from the ESKAPE group of pathogens. These also cover the list from World Health Organization's Priority pathogens such as carbapenemase-producing *Enterobacteriaceae*, methicillin-resistant *Staphylococcus aureus*, carbapenem-resistant *Klebsiella pneumoniae* [4]. Novel methods are under development for the containment of the MDR pathogen outbreaks including the usage of alternative treatments and phage therapy which may have applications in the livestock sector also [5, 6].

Comparatively fewer studies have been reported on the presence and carriage of MDR pathogens in livestock animals and their caretakers. It is also important to track the under-studied pathogens such as *Enterobacter hormaechei* present in the gastrointestinal carriage of animals and its environmental transmission to humans. *E. hormaechei* is an emerging Gram-negative pathogen of the *Enterobacteriaceae* family which can cause diverse diseases both in

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livestock and humans. As the AMR crisis can make this organism to be a problematic one, it is very essential to track the presence of *E. hormaechei* in poultry farm settings to identify its silent epidemiological dynamics in human hosts and the environment [7].

In this study, whole genome sequencing (WGS) based approach was used to characterize a previously isolated MDR *E. hormaechei* strain from the healthy broiler feces [8]. The selected organism was isolated from the farm that was providing antibiotic-amended feeds to the broiler chickens and also the isolate was resistant to most of the clinically useful antibiotics including the colistin. The WGS data suggested the strong accumulation of antibiotic resistance genes in the selected bacteria with a high probability for human pathogenicity. The detailed core genome phylogeny and comparative genome analysis has also suggested the shared features present among isolates from both human and animal sources. The results of the study suggest the immediate need for studying various emerging pathogens having zoonotic potential. Molecular screening and characterization of such emerging multidrug pathogens will help to understand the AMR evolution as well as it will be supportive to the health sector to get prepared against the infection outbreaks in a one-health manner.

Methodology

Bacterial Strain Identification and Antibiotic Sensitivity Test

The bacterial strain MS2 was isolated from the feces samples of healthy broiler chickens as described in the previous report [8]. The strain MS2 was subjected to biochemical identification using Vitek2 automated microbial typing system using VITEK® 2 GN ID card (Reference number 21341). Antibiotic sensitivity tests were conducted by the Kirby-Bauer disc diffusion method (tetracycline: 30 µg, amoxicillin/clavulanate: 20/10 µg, piperacillin/tazobactam: 100/10 µg, cefuroxime: 30 µg, cefoperazone/sulbactam: 75/10 µg, ertapenem: 10 µg, meropenem: 10 µg, ciprofloxacin: 5 µg, nitrofurantoin: 300 µg, trimethoprim-sulfamethoxazole: 1.25/23.75 µg, gentamicin: 10 µg, aztreonam: 30 µg, ceftriaxone: 10 µg) and confirmed with the VITEK- 2 AST cards as per the CLSI protocols [9, 10]. The bacterial strain was further inoculated into 10 mL of Luria Bertani broth (Himedia- M1245) and incubated for 18 h for the DNA isolation. MACHEREY–NAGEL NucleoSpin Microbial DNA Mini kit (REF-740235.50) was used for the DNA isolation as per the manufacturer's instructions and the initial quality check was done by agarose gel electrophoresis. The genomic DNA was further subjected to library preparation for whole-genome sequencing [8].

Bacterial Whole-Genome Sequencing and Analysis

The genomic DNA was used for the library preparation and paired-end sequencing was conducted by Illumina HiSeq 2500 using 151 bp read length chemistry. A total of 1 GB paired-end data was generated and read quality and preprocessing were conducted using NGSQC tool kit v-0.25.1 [11]. High quality reads having > 20 phred score were filtered and further used for the primary assembly using SPAdes v-3.14.1 to generate contigs and genome finishing was done by MeDuSa (v-2015) [12, 13]. The draft genome was further used for the gene prediction, annotation, and functional characterization using Rapid Annotation Using the Subsystem Technology annotation server (http:/rast.nmpdr.org/) [14]. The genome data was also used for the annotation of the antimicrobial resistance genes using the Resistance Gene Identifier (RGI) tool of The Comprehensive Antibiotic Resistance Database (https://card.mcmaster.ca/) [15]. The bacteriophage regions in the genome were identified using PHASTER (https://phaster.ca/) [16] and mobile genetic elements of the genome were identified using the mobile element finder tool v1.0.3 (https://cge.food.dtu.dk/services/MobileElementFinder) [17]. The probability of the strain to cause the human infection was identified using the PathogenFinder 1.1 (https://cge.food.dtu.dk/services/PathogenFinder/) [18]. The genes coding for secondary metabolites and siderophores like virulence factors were analyzed using antiSMASH V-7 beta (https://antismash.secondarymetabolites.org/) [19].

Comparative Phylogenomics

For the comparative study, a total of 82 whole-genome sequence data of *E. hormaechei* strains reported globally were collected from the genbank including human and animal origin. In the case of animal-associated strains, 8 avian strains available in the database were selected along with 24 strains associated with nonhuman mammals. The source of isolation of selected strains was from feces, blood, urine and nasal swabs as per the associated illnesses. In the case of strains isolated from humans, genome data of 50 strains were collected which were associated with conditions such as bacteremia, urinary tract infections, intra-abdominal abscess, pulmonary infections. Phylogenetic analysis and the host-specificity of *E. hormaechei* strains were conducted by SNP based core genome phylogeny approach using parsnp v1.2 [20]. The presence of antimicrobial resistance among the livestock and human origin groups was compared by using ABRicate using CARD database and a heat map was constructed using TBtools [21].

The whole-genome data of *E. hormaechei* MS2 was submitted in Genbank with an accession number GCA_020984895.1.

Results

Bacterial Strain Identification and Antibiotic Sensitivity Test

The bacterial strain MS2 selected for the study was identified as *Enterobacter* spp. by various biochemical tests was carried out through the Vitek-2 GN ID card. The MS2 was also found to be resistant to tetracycline, amoxicillin/clavulanate, piperacillin/tazobactam, cefuroxime, cefoperazone/sulbactam, ertapenem, meropenem, ciprofloxacin, nitrofurantoin, trimethoprim-sulfamethoxazole, and colistin as per the results of Kirby-Beaur disc diffusion tests. This was further confirmed by the Minimum Inhibitory Concentration data from the ViteK2 automated system. From the results, the MS2 was identified to be a multi-drug resistant organism of high concern (Table 1) as it showed resistance to the major drugs classes like Amoxicillin/clavulanate (MIC-32), Nitrofurantoin (MIC-256), Cefuroxime axetil (MIC-64), trimethoprim-sulfamethoxazole (MIC-32), carbapenems (ertapenem and meropenem MIC-8), piperacillin/tazobactam (MIC-128), cefoperazone/sulbactam (MIC-64) and colistin (MIC-16). However, the MS2 showed sensitive MIC values to amikacin (MIC-8) and tigecycline (MIC-0.5) [7].

Bacterial Whole-Genome Sequencing and Analysis

By the whole-genome sequencing, a total of 6,826,210 high-quality forward and reverse reads were obtained which were further subjected to the *denovo* assembly using SPAdes v3.14.1. The final assembly generated a total of 60 sequences having a total of 5,057,802 bases with 54.66 GC% and 4350906 bp N50. A total of 5086 genes could be predicted from the genome MS2 out of which 5004 were coding for the total proteins, 82 noncoding genes, 77tRNA genes, and 5 were for *rRNA* genes. The pubMLST analysis further demonstrated MS2 as a strain of *Enterobacter hormaechei* with 100% similarity and hence the data was submitted to Genbank as *E. hormaechei* strain MS2 (Accession no. GCA_020984895.1).

The resistome analysis using the RGI further showed the MS2 to have the presence of 7 perfect hits and 26 strict hits for the AMR genes. Among these, 13 genes were coding for the efflux pumps (*emrR, emrB, rsmA, adeF, CRP, oqxA, baeR, H-NS, KpnF, KpnE, msbA, tet(A),* and *marA*) and 11 were coding for the antibiotic inactivation (*OXA-1, AAC (6')-Ib-cr5, CTX-M-15, TEM-1, FosA2, ACT-69, ANT (3'')-IIa, catI, AAC(3)-IIe, APH(6)-Id* and *APH(3'')-Ib)* and 4 were coding for the antibiotic target alterations (*MCR-9, vanG, PBP3*, and *UhpT* (Supplementary file 1) [7].

The mobile element identifier has identified 12 different types of insertion sequences (ISKpn24, IS903, ISSen4, ISKpn43, ISKpn21, IS5075, ISEsa2, ISEc52, ISEam1, IS5, IS6100, and ISEc9) and 2 composite transposons (cn_15648_IS903 and cn_2245_IS5) from the genome data of MS2 (Supplementary file 1). IncHI2A, IncHI2, and Col440I plasmids could also be identified from the whole-genome sequence of MS2.

The MS2 genome also carried 5 intact phage regions having similarity to the phages from Gram-negative organisms such as *Escherichia* spp., *Salmonella* spp., *Erwinia* spp., and *Enterobacter* spp.

In addition, the pathogenicity prediction of MS2 has demonstrated its potential human pathogenic characteristics with a probability of 0.726 by an input proteome coverage of 0.83% and this matched the pathogenic protein family of 31 [18]. The antiSMASH results showed the presence of two siderophores such as aerobactin and enterobactin with 66% and 100% similarity (Fig. 1).

Table 1 Summary of antibiotic resistance observed for the isolate MS2 using ViteK2 automated system

Antibiotic Class	Antibiotic	MIC value	CLSI Breakpoint (mg/L)	Interpretation
Penicillins BL/BLI	Amoxicillin/Clavulanate	32	< 8	Resistant
	Piperacillin/Tazobactam	128	< 16	Resistant
Cephalosporins BL/BLI	Cefoperazone/Sulbactam	64	≤ 16	Resistant
Cephalosporins 2nd gen	Cefuroxime Axetil	64	–	Resistant
Nitrofurans	Nitrofurantoin	256	≤ 32	Resistant
Sulfonamides	Trimethoprim-Sulfamethoxazole	32	≤ 2/38	Resistant
Carbapenems	Ertapenem	8	≤ 0.5	Resistant
	Meropenem	8	≤ 1	Resistant
Aminoglycosides	Amikacin	8	≤ 16	Sensitive
Glycicylines	Tigecycline	0.5	–	Sensitive
Polymixins	Colistin	16	≤ 2	Resistant

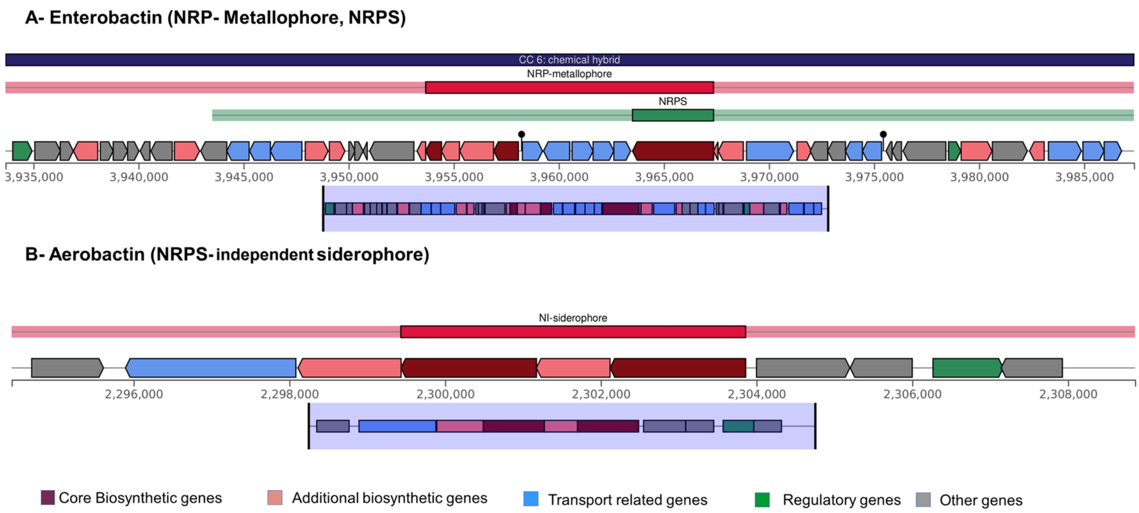

Fig. 1 Schematic representations of the biosynthetic gene clusters of aerobactin (**A**) and Enterobactin (**B**) from *E. hormaechei* MS2

Comparative Phylogenomics

The core genome-based phylogenetic analysis conducted with the clinical and livestock originated strains has identified the strain MS2 to have a versatile infective ability with both the hosts (Fig. 2). The comparative AMR gene screening has also resulted in the identification of host specific as well as shared antimicrobial resistance genes among the selected strains (Fig. 3). A total of 118 AMR genes were detected for it out of which 71 genes were responsible for antibiotic inactivation and 26 genes were for the antibiotic efflux. Seven genes were having the function of target replacement and 3 were having the potential for target alteration.

Discussion

E. hormaechei belongs to the *E. cloacae* complex group, which includes other clinically relevant pathogens like *E. kobei, E. ludwigii, E. mori, E. nimipressuralis*, and *E. bugandensis*. Among them, *E. hormaechei* has been previously identified as a zoonotic pathogen that can cause urinary tract infection, respiratory infection, and soft tissue infection in humans. However, the lack of standardized methods to identify *E. cloacae* complex group in clinical settings often leads to misidentification of pathogens and unclear treatment strategies. Furthermore, comprehensive information regarding the genomic virulence properties of this pathogen is largely unavailable in most of the virulence databases.

E. hormaechei strains have already been reported to exhibit antimicrobial resistance against various antibiotics, including piperacillin/tazobactam, cefotaxime, ceftazidime, cefoxitin, carbapenems, gentamicin, amikacin, aztreonam, tobramycin, tigecycline, ciprofloxacin, trimethoprim/sulfamethoxazole, and colistin. However, recent reports of antibiotic-resistant *E. hormaechei* strains in livestock raise the critical concerns and have implications for infectious disease treatments. A detailed analysis on the antimicrobial resistance evolution in *E. hormaechei* is essential for a comprehensive approach to combat drug resistance in the context of One Health [28].

In this study, *E. hormaechei* MS2 was isolated from the healthy broilers and found to exhibit high drug resistance. The strain displayed resistance to a range of antibiotics, including the first-line, second-line, and reserve antibiotics such as carbapenems and colistin. Subsequent whole-genome sequencing has revealed the presence of specific genes responsible for observed resistance. To the best of our knowledge, this is the first report on the presence of *MCR-9*, *FosA2,* and *ACT-69* genes in India. The *MCR-9* gene codes for the enzyme phosphoethanolamine transferase, which modifies the Lipid A structure in the outer membrane of bacteria. Similar to *MCR-1*, this gene has been reported to be plasmid mediated and is known to be transferable among bacteria. The *FosA2* enzyme belongs to the family of fosfomycin thiol transferases and it functions by inactivating the fosfomycin through the cleavage of its epoxide ring. *ACT-69* is also identified as a beta-lactamase enzyme capable of conferring resistance to a broad range of antibiotics, including the cephamycin, cephalosporin, penam, and carbapenem [22].

The mobile genetic element identifier has also identified various insertion sequences and composite transposons in MS2. Among them, the InCHI2 plasmids are commonly reported in *Salmonella* and *E. coli* and are capable of transferring the resistance against many antibiotics, including

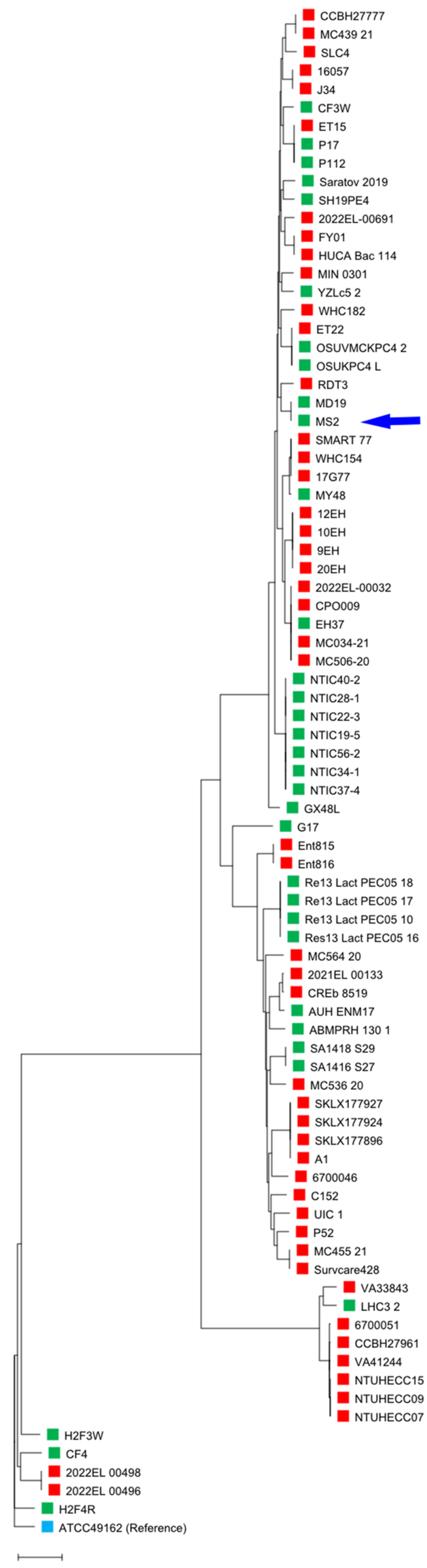

Fig. 2 SNP based core genome phylogenetic analysis of *E. hormaechei* strains isolated from livestock (red) and human hosts (green) conducted using Parsnp. ATCC 49162 was used as the reference strain (blue). The phylogenetic analysis shows the clustering of *E. hormaechei* strain MS2 together with livestock and human hosts signifying their zoonotic potential (Color figure online)

ampicillin, tetracycline, and trimethoprim. It has also been reported to carry genes for colistin resistance, such as *MCR-1*. The Col440I plasmid has a wide host range and is typically studied for its association with *QnrB* genes, which are related to fluoroquinolone resistance. The co-occurrence of *MCR-9* and the super plasmid IncHI2 in the genome of the MS2 strain indicates an increased risk of silent colistin resistance dissemination, as described in previous studies [23]. Presence of heavy deposition of phage regions also shows the MS2 to have the genomic competence to receive and disseminate antimicrobial resistance through phages at some point in its growth cycle [24]. The high pathogenicity potential of the poultry associated strain indicates the need for continuous and proper screening of zoonotic pathogens from livestock. The presence of such opportunistic pathogens in the gut of healthy broilers provide indications on its likely role to cause infections or being transmitted to the humans during the rearing and through the supply chain activities.

The siderophores are crucial for the survival and pathogenesis of bacteria as it functions as a virulence factor by scavenging the iron. Here, *E. hormaechei* MS2 was found to have two types of siderophore such as aerobactin and enterobactin. The aerobactin is a non-ribosomal protein synthase (NRPS) independent siderophore which is coded by the cluster containing 10 genes (14429nt length), while the enterobactin is coded by the NRPS gene cluster having 48 genes (53700nt). Enterobactin is the strongest siderophore present among multiple pathogens such as *Escherichia coli* and *Klebsiella pneumoniae* [25]. There are many previous reports on the antimicrobial resistance in potentially infective *E. hormaechei* strains isolated from livestock of Bangladesh showing resistance to clindamycin, sulfonamides, imipenem, streptomycin, nitrofurantoin, doxycycline, and tetracyclines [26]. However, the lack of availability of an organized dataset for the virulence factors of *Enterobacter* spp. is an important hurdle in the data analysis using WGS [27]. There are previous attempts to extract the virulence properties of the *Enterobacter* spp. through a comparative analysis with existing virulence datasets of other bacterial pathogens but the approach has inherent limitation [28]. Furthermore, the lack of an established zoonotic surveillance system especially for the opportunistic pathogens in many developing and under developing countries can misidentify the human infections transmitted through these routes and consider them as independent cases. Such

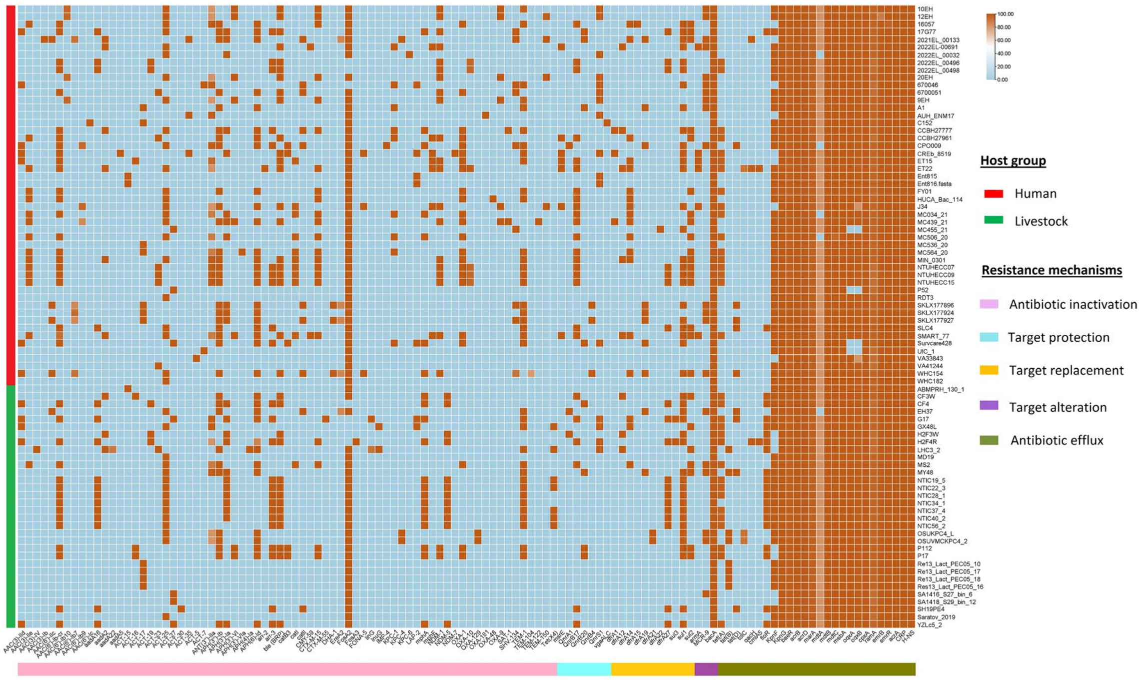

Fig. 3 Comparative genomic analysis of AMR genes in livestock and human originated strains of *E. hormaechei*. The right red bar line represents the human-associated strains and the green bar line represents the livestock-associated strains. The bottom bar line with different colors represents various types of antibiotic resistance mechanisms (Color figure online)

opportunistic infections may not be recognized as zoonotic infections unless it is present at a noticeable local outbreak. At the same time, these zoonotic opportunistic pathogens can also share the AMR genes with other susceptible organisms resulting in serious health complications.

The phylogenetic analysis of *E. hormaechei* strains, from the human and livestock originated groups, was performed using a core genome-based approach. The results of the analysis revealed the zoonotic potential of *E. hormaechei* strains, as they exhibited random clustering with both the host groups. Interestingly, *E. hormaechei* MS2 was found to share a clade with a human-associated strain, RDT3, which was isolated from the feces of a diseased human, and with MD19, which was isolated from the feces of a canine host presenting septic shock and pulmonary infection (Fig. 2). This finding further underscores the versatility of *E. hormaechei* as a zoonotic pathogen, capable of causing various types of infections in humans. However, the phylogeny also displayed certain host-specific clades, which could be attributed to the similarities in geographical locations or limited data availability used for comparisons. Many phylogenetic studies have previously been demonstrated the host-specific evolution in bacteria, such as *Photobacterium damselae*, *Edwardsiella ictalurid*, and *Edwardsiella ictaluri*. Nonetheless, the absence of such patterns in *E. hormaechei* MS2 suggests its capability to infect hosts, leading to potential zoonotic conditions [29–31].

The comparative genomic analysis of selected human and animal originated strains of *E. hormaechei* further revealed notable patterns in the global distribution of antimicrobial resistance (AMR) genes. Among the animal-associated *E. hormaechei* strains, *Tet(X4)* (Human (H)-0%/Livestock (L)-21.8%), *tetB* (H-0%/L-25%), and *NDM-5* (H-0%/L-12%) were strictly associated. Conversely, *mphA* (H-14%/L-43%), *sul1* (H-50/L-62.5%), *floR* (H-4/L-59), and *tetA* (H-34/L-50) were comparatively higher among livestock-associated strains (Fig. 3).

Furthermore, major extended-spectrum beta-lactamase (ESBL) genes, such as *CTX-M-15* (H-36/L-12.5) and *OXA-1* (H-40/L-9), were more prevalent among human-associated isolates. *TEM-1* (H-44%/L-46.87%) has showed a nearly equal distribution among both the host groups. *SHV134* had a lower prevalence among human strains and was absent in livestock-associated strains (H-18%/L-0%). At the same time, *OXA-48* was negligibly absent in both the groups. Interestingly, apart from the *MCR 9*, no other colistin resistance genes was found among the tested strains, and *MCR 9* was comparatively more prevalent among human-associated strains (H-28%/L-18.75). Similarly, *NDM-1* (H-20%/L-2%) was observed to be more common in human strains, while *NDM-5* (H-12%/L-0%) showed a strict association with livestock strains. The genes *dfrA14* (H-30%/L-9.3%) and *sul2* (H-34%/L-21%) were found to be more prevalent in human-associated strains.

Genes such as *bacA, tet(D), tolC, qacH, cmlA5, floR, KpnF, KpnG, baeR, acrB, acrD, marA, mdtA, mdtB, mdtC, msbA, oqxA, oqxB, cpxA, ramA, emrB, emrR, acrA, CRP,* and *H-NS* were present in all the tested genomes and may indicate intrinsic resistance. As there are, only limited information is available on the intrinsic resistance in *E. hormaechei* to various antibiotics such as penicillin, ampicillin, first, second, and third-generation cephalosporins (cefalexin, cefazolin, and cefotaxime), tetracyclines, carbenicillin, etc. accurate prediction of AMR is challenging. The comparative analysis of AMR in *E. hormaechei* strains revealed the global distribution of genes coding for resistance to other antibiotics such as fluoroquinolones and aminoglycosides. Immediate characterization of both intrinsic and acquired resistance genes of the *Enterobacter cloacae* complex is highly necessary [31, 32]. From the results of the analysis made, *E. hormaechei* MS2 could be identified as a multidrug resistant strain with high zoonotic infectivity. Although it is supported by the core genome phylogeny, the comparative AMR analysis suggests the need for further investigation to identify the typical host-specific nature and pattern of AMR for the easy differentiation and its origin prediction. Identification of such markers in noncore genomic regions might help to identify and tackle the zoonotic infections caused by the bacteria in the human population.

Conclusion

Zoonotic antimicrobial resistance transmission poses a significant threat to the human health and welfare. To prevent such silent spread of antimicrobial resistance (AMR), it is crucial to implement scientific monitoring and containment measures. In this study, a multidrug-resistant strain of *Enterobacter hormaechei* MS2 isolated from the feces of healthy broiler chicken was subjected to whole-genome sequencing. The detailed analysis of WGS data of MS2 was further resulted in the identification of novel genes such as MCR-9, ACT-69 and FosA2 genes for the first time from the Indian subcontinent. Hence the results of the study provide important insight into the need for a detailed surveillance through genomic analysis to get a deeper understanding on emerging AMR crisis from the zoonotic sources. Also, the findings have demonstrated substantial presence of antimicrobial resistance genes in the selected strain, indicating its potential to be a human pathogen with a high health risk. The SNP based core genome phylogenetic analysis, conducted in the study along with globally reported strains from both human and animal sources, further supported the potential of *E. hormaechei* MS2 to transmissible to humans. At the same time, certain AMR genes were found to be specific to particular host groups, which demands much scientific investigations to unravel the genetic basis of the same. The study highlights the urgent need to develop a zoonotic surveillance protocol to identify the AMR crisis with opportunistic pathogens like *E. hormaechei* especially with respect to evolution of novel resistance mechanisms.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00284-023-03492-w.

Acknowledgements Acknowledging the funding provided by The Department of Animal Husbandry, Government of Kerala, and the support form Kerala Antimicrobial Resistance Strategic Action Plan (KARSAP). Acknowledging the instrumentation facility provided by The Kerala State Council for Science, Technology and Environment (KSCSTE) SRS project.

Author Contributions SS: conceptualization, data analysis, writing; MP: Data analysis, writing; PRP: review and editing, CCN: data analysis, NP: review and editing; SSA: review and editing, JM: supervision and editing; EKRR: conceptualization, supervision and editing.

Funding The Department of Animal Husbandry, Government of Kerala.

Data availability The datasets analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest This is to declare that the authors have no conflict of interest on this research work.

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